CPTB1: Drinking Water Treatment Facility

CPFD Key

- **Interactor (CI/PI/CPI)**
- **Trust Boundary (CTB/PTB/CPTB)**
- **Store(CS/PS/CPS)**
- **Nested Store (NS)**
- **Flow (CF/PF/CPF)**
- **Link (CL/PL/CPL)**
- **Process (CP/PP/CPP)**
- **Device (CD/PD/CPD)**

PI2: Landfill

CPI2: Electric Utility Provider

CPF7: Electricity

CPD13: Main Power Distribution Panel

PTB2: Electrical shed

PS4: Post Chemical

PF17: Chemical

PF17: Treated Water

CPP6: Chemical Post-treatment

PF15: Disinfected Water

PD4: Disinfection Tank

CPP5: Disinfection

PL6: Filtered Water Pipe

PF12: Filtered Water

PD3: Filtration Tank

CPP4: Filtration

PF19: Plant Waste

PF11: Filter residue

CPP7: Waste collection and disposal

PF16: Mechanical

CPF6: Telemetry

CPD11: PLC Actuators (Feed pumps, dosing valves)

CPD12: RTU Sensors

PTB1: Post-treatment shed

PF14: Mechanical

CPF5: Telemetry

CPD10: PLC Actuators (Feed pumps, UV lamps, dosing valves)

CPD9: RTU Sensors (Chlorine, ORP, UV, pH, flow)

PF13: Chemical

PS3: Chemical Disinfectant

PF10: Mechanical

CPF4: Telemetry

CPD8: PLC Actuators (filters, media, backwash)

CPD7: RTU Sensors (Turbidity, ΔP, flow, particle)

PD5: Storage Pump

PF18: Treated Water

PS5: Water Storage

PF19: Treated Water

PI3: Water Distribution Network

CF4: Remote Telemetry & Control

CI1: Org Intranet

CF3: Remote Telemetry & Control

CD1: Internet Link / Firewall

CF2: Remote Telemetry & Control

CPD14: SCADA Computer / HMI

CF1: Telemetry & Control

CPI1: Onsite Human Plant Operator(s)

CF5: Remote Telemetry & Control

CI2: Internet

CPI3: Offsite Human Plant Operator(s)

CPF15: SCADA

CPF14: SCADA

CPF13: SCADA

CPF12: SCADA

CPF8: SCADA

CPF9: SCADA

CPF10: SCADA

CPF11: SCADA

CPD1: RTU Sensors (Level, flow, turbidity, temp, conductivity, pH)

PS1: Coagulant / Flocculant

CPD3: PLC Actuators (mixers, dosing pumps, valves)

CPD4: RTU Sensors (pH, turbidity, streaming current)

CPD5: RTU Sensors (Sludge blanket, turbidity, flow)

CPD6: PLC Actuators (clarifiers, sludge scrapers, tube settlers)

PS2: Sediment

PF7: Clarified Water

PL4: Clarified Water Pipe

PF9: Sediment

CPF1: Telemetry

PF3: Chemical

PF4: Mechanical

CPF2: Telemetry

CPF3: Telemetry

PF6: Mechanical

PI1: Water Source

PF1: Source Water

CPP1: Intake

PF2: Source Water

CPP2: Coagulation / Flocculation

PF5: Flocculated Water

CPP3: Sedimentation

PF8: Sludge

PL5: Sludge Pipe

PD1: Coagulation / Flocculation Tank

PD2: Sedimentation Tank

PF20: Mechanical

PL1: Source Water Intake Pipe

CPD2: PLC Actuators (gates, screens, pumps)

PL2: Source Water Transport Pipe

PL3: Flocculated Water Pipe